Appendix: Integration

The Riemann integral, multiple and improper integrals

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A.1 Definition of the Riemann integral

Motivation. The whole book uses only the Riemann integral. This appendix records its definition and basic properties, the approximation of integrable functions by continuous ones (used for Fourier series of integrable functions), and Lebesgue's criterion: a bounded function is integrable exactly when its discontinuities form a set of measure zero.

Let f be a bounded real-valued function on $[a,b]$. A **partition** P of $[a,b]$ is a finite sequence $a = x_0 < x_1 < \cdots < x_N = b$; let $I_j = [x_{j-1}, x_j]$ and $|I_j| = x_j - x_{j-1}$. The upper and lower sums are

$$\mathcal{U}(P,f) = \sum_{j=1}^{N} \Big[\sup_{x\in I_j} f(x)\Big]|I_j|, \qquad \mathcal{L}(P,f) = \sum_{j=1}^{N} \Big[\inf_{x\in I_j} f(x)\Big]|I_j|.$$

A partition P' is a **refinement** of P if $P' \supset P$. Adding points does not increase $\mathcal{U}$ nor decrease $\mathcal{L}$: if P' refines P,

$$\mathcal{L}(P,f) \le \mathcal{L}(P',f) \le \mathcal{U}(P',f) \le \mathcal{U}(P,f).$$

(Splitting I_j into two pieces replaces $\sup_{I_j} f \cdot |I_j|$ by a sum of two terms with smaller or equal sups.) Hence for any P_1, P_2, the common refinement $P' = P_1 \cup P_2$ gives $\mathcal{L}(P_2,f) \le \mathcal{L}(P',f) \le \mathcal{U}(P',f) \le \mathcal{U}(P_1,f)$: every lower sum is $\le$ every upper sum. With $\mathcal{U} = \inf_P \mathcal{U}(P,f)$ and $\mathcal{L} = \sup_P \mathcal{L}(P,f)$ we get $\mathcal{L} \le \mathcal{U}$.

Definition

f is **(Riemann) integrable** if for every $\epsilon > 0$ there is a partition P with $\mathcal{U}(P,f) - \mathcal{L}(P,f) < \epsilon$. Then $\mathcal{U} = \mathcal{L}$, and this common value is $\int_a^b f(x)\,dx$. For complex $f = u + iv$, f is integrable if u and v are, and $\int f = \int u + i\int v$.

Equivalently, with Riemann sums $\mathcal{R}(P,f) = \sum f(x_j^*)|I_j|$ $(x_j^* \in I_j)$ and mesh $\max |I_j|$, f is integrable iff $\mathcal{R}(P,f)$ converges as the mesh tends to 0. Continuous functions are integrable: by uniform continuity, for small mesh $\sup_{I_j} f - \inf_{I_j} f < \epsilon$ on every I_j, so $\mathcal{U} - \mathcal{L} \le \epsilon(b-a)$.

A.1.1 Basic properties

Proposition 1.1

If f and g are integrable on $[a,b]$, then:
(i) $f+g$ is integrable, and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.
(ii) If $c \in \mathbb{C}$, then cf is integrable and $\int_a^b cf = c\int_a^b f$.
(iii) If f, g are real-valued and $f \le g$, then $\int_a^b f \le \int_a^b g$.
(iv) If $c \in [a,b]$, then f is integrable on $[a,c]$ and $[c,b]$, and $\int_a^b f = \int_a^c f + \int_c^b f$.

Proof. (i) Assume f, g real. For any P, $\mathcal{U}(P,f+g) \le \mathcal{U}(P,f) + \mathcal{U}(P,g)$ and $\mathcal{L}(P,f+g) \ge \mathcal{L}(P,f) + \mathcal{L}(P,g)$, since the sup of a sum is at most the sum of the sups. Given ϵ, take P_1, P_2 with $\mathcal{U} - \mathcal{L} < \epsilon$ for f and g respectively, and a common refinement P_0. Then $\mathcal{U}(P_0, f+g) - \mathcal{L}(P_0, f+g) < 2\epsilon$, so $f+g$ is integrable. Its integral I satisfies

$$I \le \mathcal{U}(P_0, f+g) \le \mathcal{U}(P_0,f) + \mathcal{U}(P_0,g) \le \int f + \int g + 2\epsilon,$$

and similarly $I \ge \int f + \int g - 2\epsilon$ from the lower sums.

(ii) For real $c \ge 0$, $\mathcal{U}(P,cf) = c\,\mathcal{U}(P,f)$ and $\mathcal{L}(P,cf) = c\,\mathcal{L}(P,f)$. For $c = -1$, $\mathcal{U}(P,-f) = -\mathcal{L}(P,f)$ and $\mathcal{L}(P,-f) = -\mathcal{U}(P,f)$. Complex c then follows by splitting into real and imaginary parts, using (i).

(iii) $g - f \geq 0$ gives $\mathcal{L}(P, g - f) \geq 0$, so $\int (g - f) \geq 0$; apply (i) and (ii).

(iv) Refining by the point c, we may take partitions containing c; then $\mathcal{U}(P, f) = \mathcal{U}(P', f) + \mathcal{U}(P'', f)$ with P', P'' the induced partitions of $[a, c]$ and $[c, b]$, and the same for $\mathcal{L}$. Hence $\mathcal{U} - \mathcal{L}$ on each piece is $< \epsilon$, and the integrals add. ■

Consequences: if f is integrable then so is $|f|$, and $\left|\int f\right| \leq \int |f|$ (Lemma 1.2 with $\varphi(x) = |x|$; for complex f use $\varphi(u, v)$ on the bounded range, and write $\int f = e^{i\theta} \left|\int f\right|$, so $\left|\int f\right| = \int \mathrm{Re}(e^{-i\theta} f) \leq \int |f|$). Also the product of two integrable functions is integrable, since $fg = \frac{1}{4}[(f + g)^2 - (f - g)^2]$ and squares are integrable by Lemma 1.2.

Lemma 1.2

If f is real-valued and integrable on $[a, b]$, and φ is a real-valued continuous function on $\mathbb{R}$, then $\varphi \circ f$ is integrable on $[a, b]$.

Proof. Let $\epsilon > 0$ and $|f| \leq M$. Since φ is uniformly continuous on $[-M, M]$, choose $\delta > 0$ (with $\delta < \epsilon$) so that $|y_1 - y_2| < \delta$ implies $|\varphi(y_1) - \varphi(y_2)| < \epsilon$. Choose P with $\mathcal{U}(P, f) - \mathcal{L}(P, f) < \delta^2$. Let I_j be the subintervals of P, and split the indices into Λ, where $\sup_{I_j} f - \inf_{I_j} f < \delta$, and Λ', the rest. On I_j with $j \in \Lambda$, the oscillation of $\varphi \circ f$ is $\leq \epsilon$. For $j \in \Lambda'$,

$$\delta \sum_{j \in \Lambda'} |I_j| \leq \sum_{j \in \Lambda'} \Big(\sup_{I_j} f - \inf_{I_j} f \Big) |I_j| < \delta^2,$$

so $\sum_{\Lambda'} |I_j| < \delta$. With $B = \sup_{[-M,M]} |\varphi|$,

$$\mathcal{U}(P, \varphi \circ f) - \mathcal{L}(P, \varphi \circ f) \leq \epsilon(b - a) + 2B\delta < \epsilon(b - a + 2B).$$

■

Proposition 1.3

A bounded monotonic function f on $[a, b]$ is integrable.

Proof. We may take $a = 0$, $b = 1$ and f increasing. For the uniform partition P_N with $x_j = j/N$,

$$\mathcal{U}(P_N, f) - \mathcal{L}(P_N, f) = \sum_{j=1}^{N} \frac{1}{N} [f(x_j) - f(x_{j-1})] = \frac{f(1) - f(0)}{N} \to 0.$$

■

Proposition 1.4

Let f be a bounded function on $[a, b]$. If $c \in (a, b)$, and for all small $\delta > 0$ the function f is integrable on $[a, c - \delta]$ and $[c + \delta, b]$, then f is integrable on $[a, b]$.

Proof. Let $|f| \leq M$ and $\epsilon > 0$. Choose $\delta > 0$ with $4\delta M \leq \epsilon/3$, then partitions P_1 of $[a, c - \delta]$ and P_2 of $[c + \delta, b]$ with $\mathcal{U} - \mathcal{L} < \epsilon/3$ on each. Let $P = P_1 \cup P_2$, a partition of $[a, b]$ that has $[c - \delta, c + \delta]$ as one subinterval. Then

$$\mathcal{U}(P, f) - \mathcal{L}(P, f) \leq (\mathcal{U}(P_1, f) - \mathcal{L}(P_1, f)) + 2M \cdot 2\delta + (\mathcal{U}(P_2, f) - \mathcal{L}(P_2, f)) < \epsilon.$$

■

By induction, a bounded function on $[a, b]$ that is continuous except at finitely many points is integrable.

A function on the circle (equivalently, a 2π-periodic function on $\mathbb{R}$) is **integrable** if it is integrable on every finite interval, i.e. on $[-\pi, \pi]$.

Lemma 1.5

Suppose f is integrable on the circle and $|f| \le B$. Then there is a sequence $\{f_k\}$ of continuous functions on the circle with

$$\sup_{x\in[-\pi,\pi]} |f_k(x)| \le B \quad \text{for all } k, \qquad \int_{-\pi}^{\pi} |f(x) - f_k(x)|\,dx \to 0 \quad \text{as } k \to \infty.$$

Proof. Assume f real (otherwise treat the real and imaginary parts separately; the bound becomes $2B$, and one can then radially truncate to B without increasing the error, since $z \mapsto$ the nearest point of $\{|w| \le B\}$ is 1-Lipschitz).

Step Step function. Given ϵ, choose a partition $-\pi = x_0 < \cdots < x_N = \pi$ with $\mathcal{U} - \mathcal{L} < \epsilon$, and let $f^*(x) = \sup_{x_{j-1}\le y\le x_j} f(y)$ for $x \in [x_{j-1}, x_j)$. Then $|f^*| \le B$ and

$$\int_{-\pi}^{\pi} |f^* - f| = \int_{-\pi}^{\pi} (f^* - f) \le \mathcal{U} - \mathcal{L} < \epsilon.$$

Step Smoothing the jumps. For small $\delta > 0$ let $\tilde{f} = f^*$ at distance $\ge \delta$ from every x_j. Near x_j $(1 \le j \le N-1)$ let $\tilde{f}$ be linear on $[x_j - \delta, x_j + \delta]$ with $\tilde{f}(x_j \pm \delta) = f^*(x_j \pm \delta)$. Near $-\pi$, let $\tilde{f}$ be linear on $[-\pi, -\pi + \delta]$ with $\tilde{f}(-\pi) = 0$, and similarly near π. Then $\tilde{f}(-\pi) = \tilde{f}(\pi) = 0$, so $\tilde{f}$ extends to a continuous 2π-periodic function with $|\tilde{f}| \le B$. It differs from f^* only on N intervals of length $\le 2\delta$, so $\int |f^* - \tilde{f}| \le 2B \cdot N \cdot 2\delta < \epsilon$ for δ small.

Hence $\int |f - \tilde{f}| < 2\epsilon$. Take $\epsilon = 1/k$ and $f_k = \tilde{f}$. ∎

A.1.2 Sets of measure zero and discontinuities of integrable functions

A set $E \subset \mathbb{R}$ has **measure** 0 if for every $\epsilon > 0$ there are countably many open intervals I_k with $E \subset \bigcup I_k$ and $\sum |I_k| < \epsilon$. Countable sets have measure 0 (cover the k-th point by an interval of length $\epsilon 2^{-k}$). If E is compact and has measure 0, finitely many of the I_k already cover E.

Lemma 1.6

The union of countably many sets of measure 0 has measure 0.

Proof. Let $E = \bigcup_i E_i$. Given ϵ, cover E_i by intervals $\{I_{k,i}\}_k$ with $\sum_k |I_{k,i}| < \epsilon/2^i$. Then $\{I_{k,i}\}_{k,i}$ is a countable cover of E with total length $< \sum_i \epsilon 2^{-i} = \epsilon$. ∎

Theorem 1.7 (Lebesgue)

A bounded function f on $[a, b]$ is integrable if and only if its set of discontinuities has measure 0.

Let $J = [a, b]$ and $I(c, r) = (c - r, c + r)$. The **oscillation** of f on $I(c, r)$ is $\operatorname{osc}(f, c, r) = \sup |f(x) - f(y)|$ over $x, y \in J \cap I(c, r)$, and the oscillation at c is $\operatorname{osc}(f, c) = \lim_{r\to 0} \operatorname{osc}(f, c, r)$ (a decreasing limit of non-negative numbers). Clearly f is continuous at c iff $\operatorname{osc}(f, c) = 0$. For $\epsilon > 0$ let $A_\epsilon = \{c \in J : \operatorname{osc}(f, c) \ge \epsilon\}$. Then the set of discontinuities is $D = \bigcup_{\epsilon>0} A_\epsilon = \bigcup_n A_{1/n}$.

Lemma 1.8

For $\epsilon > 0$, the set A_ϵ is closed, hence compact.

Proof. Suppose $c_n \in A_\epsilon$, $c_n \to c$, and $c \notin A_\epsilon$. Write $\operatorname{osc}(f, c) = \epsilon - \delta$ with $\delta > 0$. Choose r with $\operatorname{osc}(f, c, r) < \epsilon - \delta/2$ and n with $|c_n - c| < r/2$. Since $I(c_n, r/2) \subset I(c, r)$, $\operatorname{osc}(f, c_n, r/2) < \epsilon$, so $\operatorname{osc}(f, c_n) < \epsilon$, a contradiction. ∎

Proof of Theorem 1.7. Let $|f| \le M$.

Step Measure zero ⇒ integrable. Let $\epsilon > 0$. Since $A_\epsilon \subset D$ is compact with measure 0, it is covered by finitely many open intervals $I_1, \dots, I_N$ of total length $< \epsilon$. The complement in J of their union is compact, and each of its points z has an open interval F_z with $\sup_{x,y \in J \cap F_z} |f(x) - f(y)| \le \epsilon$ (since $z \notin A_\epsilon$, after shrinking). Choose a finite subcover $I_{N+1}, \dots, I_{N'}$ of these. The endpoints of $I_1, \dots, I_{N'}$ in J give a partition P of J. Each of its subintervals lies in some I_k: in the closure of one with $k \le N$ (contributing at most $2M$ times length) or in one with $k > N$ (oscillation $\le \epsilon$). Hence

$$\mathcal{U}(P, f) - \mathcal{L}(P, f) \le 2M \sum_{j=1}^{N} |I_j| + \epsilon(b - a) \le (2M + b - a)\epsilon.$$

Step Integrable ⇒ measure zero. By Lemma 1.6 it suffices to show each $A_{1/n}$ has measure 0. Given ϵ, take $P = \{x_0, \dots, x_N\}$ with $\mathcal{U}(P, f) - \mathcal{L}(P, f) < \epsilon/n$. If $A_{1/n}$ meets the open interval $I_j = (x_{j-1}, x_j)$, then $\sup_{I_j} f - \inf_{I_j} f \ge 1/n$ there, so

$$\frac{1}{n} \sum_{j:\, I_j \cap A_{1/n} \neq \emptyset} |I_j| \le \mathcal{U}(P, f) - \mathcal{L}(P, f) < \frac{\epsilon}{n}.$$

Slightly enlarging these intervals, and adding tiny intervals around the finitely many points x_j, covers $A_{1/n}$ by open intervals of total length $\le 2\epsilon$. ∎

(Incidentally this gives another proof that fg is integrable when f and g are: the discontinuities of fg lie in the union of those of f and g.)

A.2 Multiple integrals

Motivation. Integrals over rectangles in $\mathbb{R}^d$ are defined exactly as in one variable. For continuous functions we need three facts: iterated integration (Fubini), the change of variables formula with the Jacobian factor $|\det Dg|$, and its special case, spherical coordinates.

A.2.1 The Riemann integral in $\mathbb{R}^d$

A closed rectangle is $R = [a_1, b_1] \times \cdots \times [a_d, b_d]$. If P_j is a partition of $[a_j, b_j]$, then $P = (P_1, \dots, P_d)$ is a partition of R, whose subrectangles $S = S_1 \times \cdots \times S_d$ have volume $|S| = |S_1| \cdots |S_d|$. Upper and lower sums

$$\mathcal{U}(P, f) = \sum_S \Big[\sup_S f\Big]|S|, \qquad \mathcal{L}(P, f) = \sum_S \Big[\inf_S f\Big]|S|,$$

refinements, integrability and $\int_R f(x)\,dx$ are defined exactly as in one dimension, with the same monotonicity under refinement. Continuous functions on R are integrable, by uniform continuity. If f is continuous on a closed ball B (or on a set whose boundary has measure zero in the d-dimensional sense), let $g = f$ on B and $g = 0$ outside. Then g is integrable on any rectangle $R \supset B$, its discontinuities lying on ∂B, and $\int_B f = \int_R g$ is independent of R.

A.2.2 Repeated integrals

Theorem 2.1 (Fubini, continuous case)

Let f be continuous on a closed rectangle $R = R_1 \times R_2 \subset \mathbb{R}^d$, $R_i \subset \mathbb{R}^{d_i}$, $d = d_1 + d_2$. Writing $x = (x_1, x_2)$, the function $F(x_1) = \int_{R_2} f(x_1, x_2)\, dx_2$ is continuous on R_1, and

$$\int_R f(x)\, dx = \int_{R_1} \Big(\int_{R_2} f(x_1, x_2)\, dx_2 \Big) dx_1.$$

Proof.

Step Continuity. $|F(x_1) - F(x_1')| \le \int_{R_2} |f(x_1, x_2) - f(x_1', x_2)|\, dx_2 \le |R_2| \sup_{x_2} |f(x_1, x_2) - f(x_1', x_2)|$, which is small when $|x_1 - x_1'|$ is small, by uniform continuity of f on R.

Step Identity. Let P_1, P_2 be partitions of R_1, R_2, with subrectangles S and T. Then

$$\sup_{S\times T} f \ge \sup_{x_1 \in S} \Big(\sup_{x_2 \in T} f(x_1, x_2) \Big), \qquad \inf_{S\times T} f \le \inf_{x_1 \in S} \Big(\inf_{x_2 \in T} f(x_1, x_2) \Big).$$

For each x_1, $\mathcal{L}(P_2, f(x_1, \cdot)) \le F(x_1) \le \mathcal{U}(P_2, f(x_1, \cdot))$, i.e.

$$\sum_T \Big[\inf_{x_2 \in T} f(x_1, x_2) \Big] |T| \le F(x_1) \le \sum_T \Big[\sup_{x_2 \in T} f(x_1, x_2) \Big] |T|.$$

Taking $\sup_{x_1 \in S}$ of the right side and using the first inequality,

$$\mathcal{U}(P_1, F) \le \sum_S \sum_T \Big[\sup_{S\times T} f \Big] |T||S| = \mathcal{U}(P, f),$$

and likewise $\mathcal{L}(P, f) \le \mathcal{L}(P_1, F)$, for $P = (P_1, P_2)$. Since f is integrable, $\mathcal{U}(P, f) - \mathcal{L}(P, f)$ can be made arbitrarily small, and so $\int_{R_1} F = \int_R f$. ∎

By induction, $\int_R f = \int_{a_1}^{b_1} \cdots \int_{a_d}^{b_d} f\, dx_d \cdots dx_1$, in any order.

A.2.3 The change of variables formula

A map $g : A \to B$ between compact sets is a C^1 **diffeomorphism** if it is a bijection that extends to a C^1 map on an open neighbourhood of A, its inverse likewise extends to a C^1 map near B, and $\det Dg \neq 0$ on A.

Theorem 2.2 (change of variables)

Suppose A and B are compact subsets of $\mathbb{R}^d$ and $g : A \to B$ is a C^1 diffeomorphism. If f is continuous on B, then

$$\int_{g(A)} f(x)\, dx = \int_A f(g(y))\, |\det(Dg)(y)|\, dy.$$

Proof. [Supplement] We assume (as is implicit in the statement) that A and B have boundaries of measure zero, so both integrals are defined. Write $\mathrm{vol}(E) = \int_E 1$ for such sets E.

Step Linear maps. If L is invertible linear and R is a rectangle, then $\mathrm{vol}(L(R)) = |\det L|\, \mathrm{vol}(R)$. Both sides are multiplicative in L (the left because $\mathrm{vol}(L(E)) = c_L \mathrm{vol}(E)$ for all such E, by approximating E with rectangles). So it suffices to check this for elementary matrices. A permutation of coordinates or a scaling $x_i \mapsto c x_i$ maps rectangles to rectangles with the right volume. For a shear $x_1 \mapsto x_1 + t x_2$, Fubini (Theorem 2.1) shows that each slice in x_1 is just translated, so volume is preserved and $\det = 1$.

Step Local estimate. Fix $\epsilon \in (0,1)$. By uniform continuity of Dg and $(Dg)^{-1}$ on A, there is $\eta > 0$ such that for every cube Q in A of side $< \eta$ with centre a,

$$g(a) + Dg(a)\big((1-\epsilon)(Q-a)\big) \subset g(Q) \subset g(a) + Dg(a)\big((1+\epsilon)(Q-a)\big). \tag{A.1}$$

Indeed, $h = Dg(a)^{-1}(g - g(a))$ satisfies $|h(y)-(y-a)| \le \epsilon'|y-a|$ near a by the mean value inequality, with ϵ' uniform in a. Such an h maps $Q-a$ into $(1+\epsilon)(Q-a)$. Since h is a homeomorphism onto its image and moves the boundary of Q by less than the gap, the image covers $(1-\epsilon)(Q-a)$. By the linear case and (A.1),

$$(1-\epsilon)^d |\det Dg(a)|\,|Q| \le \operatorname{vol}(g(Q)) \le (1+\epsilon)^d |\det Dg(a)|\,|Q|.$$

Step Summation. Assume first $f \ge 0$; general f follows by linearity, writing f as a difference of non-negative continuous functions plus a constant. Partition a rectangle containing A into cubes Q_k of side $< \eta$. Let $\mathcal{K}$ be the cubes contained in A; the others meet ∂A and have total volume $\to 0$ as $\eta \to 0$, and so do their images, since g is Lipschitz. The images $g(Q_k)$ overlap only on boundaries, which have measure zero, so

$$\int_{g(A)} f = \sum_{k\in\mathcal{K}} \int_{g(Q_k)} f + o(1) = \sum_{k\in\mathcal{K}} f(g(a_k))\operatorname{vol}(g(Q_k)) + o(1),$$

with a_k the centre of Q_k, using uniform continuity of f. By the local estimate this lies within a factor $(1\pm\epsilon)^d$ of $\sum_k f(g(a_k))|\det Dg(a_k)||Q_k|$, which is a Riemann sum for $\int_A f(g(y))|\det Dg(y)|dy$. Letting $\eta \to 0$ and then $\epsilon \to 0$ gives the formula. ∎

The factor $|\det Dg|$ is the new infinitesimal volume element: for linear $g = L$, $|L(R)| = |\det L|\,|R|$.

A.2.4 Spherical coordinates

Spherical coordinates in $\mathbb{R}^d$ are $x = g(r, \theta_1, \dots, \theta_{d-1})$ with

$$\begin{aligned} x_1 &= r\sin\theta_1 \sin\theta_2 \cdots \sin\theta_{d-2}\cos\theta_{d-1},\\ x_2 &= r\sin\theta_1 \sin\theta_2 \cdots \sin\theta_{d-2}\sin\theta_{d-1},\\ &\vdots\\ x_{d-1} &= r\sin\theta_1\sin\theta_2, \qquad x_d = r\cos\theta_1, \end{aligned}$$

where $0 \le \theta_i \le \pi$ for $1 \le i \le d-2$ and $0 \le \theta_{d-1} \le 2\pi$. The Jacobian determinant is $r^{d-1}\sin^{d-2}\theta_1 \sin^{d-3}\theta_2 \cdots \sin\theta_{d-2}$. Every $x \ne 0$ is uniquely $x = r\gamma$ with $r > 0$ and $\gamma \in S^{d-1}$. Define

$$\int_{S^{d-1}} f(\gamma)\, d\sigma(\gamma) = \int_0^\pi \cdots \int_0^\pi \int_0^{2\pi} f(g(1,\theta)) \sin^{d-2}\theta_1 \cdots \sin\theta_{d-2}\, d\theta_{d-1}\cdots d\theta_1.$$

Then the change of variables formula gives, for the ball $B(0,N)$,

$$\int_{B(0,N)} f(x)\,dx = \int_{S^{d-1}} \int_0^N f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma). \tag{A.2}$$

(The map g fails to be a diffeomorphism on a set of measure zero, which is handled by applying Theorem 2.2 on $\{\delta \le r,\ \delta \le \theta_i \le \pi - \delta\}$ and letting $\delta \to 0$.) The area of the unit sphere is $\omega_d = \int_{S^{d-1}} d\sigma$.

For the annulus $A(R_1,R_2) = \{R_1 \le |x| \le R_2\}$ and $\lambda \in \mathbb{R}$, (A.2) gives

$$\int_{A(R_1,R_2)} |x|^\lambda\, dx = \omega_d \int_{R_1}^{R_2} r^{\lambda+d-1} dr = \begin{cases} \dfrac{\omega_d}{\lambda+d}(R_2^{\lambda+d} - R_1^{\lambda+d}), & \lambda \ne -d,\\ \omega_d \log(R_2/R_1), & \lambda = -d. \end{cases}$$

Consequences: the integral of $|x|^\lambda$ over the ball $|x| \le 1$ is finite as $R_1 \to 0$ iff $\lambda > -d$, and over $|x| \ge 1$ it is finite as $R_2 \to \infty$ iff $\lambda < -d$.

A.3 Improper integrals. Integration over $\mathbb{R}^d$

Motivation. Fourier transforms require integrals over all of $\mathbb{R}^d$. For continuous functions of moderate decrease the integral is a limit over growing cubes (or balls); Fubini and spherical coordinates extend to this setting.

A.3.1 Integration of functions of moderate decrease

Let $Q_N = \{|x_j| \le N/2,\ 1 \le j \le d\}$. For continuous f on $\mathbb{R}^d$, if $\lim_{N\to\infty} \int_{Q_N} f$ exists we denote it by $\int_{\mathbb{R}^d} f(x)\,dx$. A continuous f is of **moderate decrease** if

$$|f(x)| \le \frac{A}{1+|x|^{d+1}} \quad \text{for some } A > 0.$$

(For $d = 1$ this is the definition of Chapter 5; the Poisson kernel $\mathcal{P}_y(x) = \frac{1}{\pi}\frac{y}{x^2+y^2}$ is an example.)

For such f the limit exists. Let $I_N = \int_{Q_N} f$. For $M > N$, $Q_M - Q_N$ lies in the annulus $A(aN, bM)$ with $a = 1/2$ and $b = \sqrt{d}/2$, since $\{|x| < N/2\} \subset Q_N \subset \{|x| \le \sqrt{d}N/2\}$. So, by the annulus formula with $\lambda = -d-1$,

$$|I_M - I_N| \le A \int_{aN \le |x| \le bM} |x|^{-d-1} dx = A\omega_d\Big(\frac{1}{aN} - \frac{1}{bM}\Big) \le \frac{C}{N}.$$

Thus $\{I_N\}$ is Cauchy. The same argument shows that $\lim_N \int_{B_N} f$ over balls exists and has the same value, since $|\int_{Q_N} f - \int_{B_{N/2}} f| \le C/N$.

The integral over $\mathbb{R}^d$ is linear, and is invariant under translations and rotations: $\int f(x+h)\,dx = \int f(x)\,dx$, and $\int f(R(x))\,dx = \int f(x)\,dx$ for rotations R. It also satisfies $\int f(\delta x)\,dx = \delta^{-d}\int f(x)\,dx$ for $\delta > 0$. Each follows from the corresponding statement on cubes or balls (Theorem 2.2, with $|\det| = 1$ or δ^d) by letting $N \to \infty$; translates of Q_N differ from Q_N by a set where $|f|$ contributes $O(1/N)$.

A.3.2 Repeated integrals

Theorem 3.1

Suppose f is continuous on $\mathbb{R}^2$ and of moderate decrease. Then $F(x_1) = \int_{\mathbb{R}} f(x_1, x_2)\,dx_2$ is of moderate decrease on $\mathbb{R}$, and

$$\int_{\mathbb{R}^2} f(x)\,dx = \int_{\mathbb{R}} \Big(\int_{\mathbb{R}} f(x_1, x_2)\,dx_2 \Big) dx_1.$$

Proof.

Step Decrease of F. With $|f(x)| \le A/(1+|x|^3)$, split $|F(x_1)| \le \int_{|x_2|\le|x_1|} + \int_{|x_2|\ge|x_1|}$ of $A\,dx_2/(1+(x_1^2+x_2^2)^{3/2})$. On the first region the integrand is $\le A/(1+|x_1|^3)$, so that part is $\le 2A|x_1|/(1+|x_1|^3) \le A'/(1+|x_1|^2)$. On the second, the integrand is $\le A/(1+|x_2|^3)$, so that part is $\le A''\int_{|x_2|\ge|x_1|} dx_2/(1+|x_2|^3) \le A'''/(1+|x_1|^2)$. F is continuous: the integrals over $|x_2| \le N$ are continuous (Theorem 2.1) and converge uniformly in x_1, the tails being $\le \int_{|x_2|>N} A/(1+|x_2|^3)$.

Step Identity. Given $\epsilon > 0$, choose N so large that $|\int_{\mathbb{R}^2} f - \int_{Q_N} f| < \epsilon$. Also choose it so that $\int_{|x_1|>N/2} |F| < \epsilon$, and so that $\int_{|x_1|\le N/2}\int_{|x_2|>N/2} |f|\,dx_2\,dx_1 < \epsilon$; the last is possible since the inner integral is $\le \min\big(\frac{C}{1+|x_1|^2}, \frac{C}{N}\big)$. By Theorem 2.1 on Q_N,

$$\int_{Q_N} f = \int_{|x_1|\le N/2}\int_{|x_2|\le N/2} f\,dx_2\,dx_1,$$

which differs from $\int_{|x_1|\le N/2} F(x_1)dx_1$ by $< \epsilon$, which in turn differs from $\int_{\mathbb{R}} F$ by $< \epsilon$. Hence $|\int_{\mathbb{R}^2} f - \int_{\mathbb{R}} F| < 3\epsilon$. ■

The same holds in $\mathbb{R}^d = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ with the same proof.

A.3.3 Spherical coordinates

If f is of moderate decrease on $\mathbb{R}^d$, then for fixed $\gamma \in S^{d-1}$,

$$|f(r\gamma)r^{d-1}| \le \frac{Ar^{d-1}}{1+r^{d+1}} \le \frac{B}{1+r^2},$$

so $r \mapsto f(r\gamma)r^{d-1}$ is of moderate decrease on $[0,\infty)$. Letting $N \to \infty$ in (A.2) (the convergence of the inner integral is uniform in γ) gives

$$\int_{\mathbb{R}^d} f(x)\,dx = \int_{S^{d-1}} \int_0^\infty f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma).$$

Combining this with rotation invariance $\int_{\mathbb{R}^d} f(R(x))\,dx = \int_{\mathbb{R}^d} f(x)\,dx$ yields rotation invariance of the surface measure:

$$\int_{S^{d-1}} f(R(\gamma))\,d\sigma(\gamma) = \int_{S^{d-1}} f(\gamma)\,d\sigma(\gamma) \tag{A.3}$$

for continuous f on S^{d-1}. (Apply the previous formula to $F(x) = f(x/|x|)\varphi(|x|)$ with $\varphi \ge 0$ continuous, compactly supported in $(0,\infty)$, $\int_0^\infty \varphi(r)r^{d-1}dr = 1$.)